

The Bovine Basic Pancreatic Trypsin Inhibitor (Kunitz Inhibitor): A Milestone Protein^π

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Abstract: The pancreatic Kunitz inhibitor, also known as aprotinin, bovine basic pancreatic trypsin inhibitor (BPTI), and trypsin-kallikrein inhibitor, is one of the most extensively studied globular proteins. It has proved to be a particularly attractive and powerful tool for studying protein conformation as well as molecular bases of protein/protein interaction(s) and (macro)molecular recognition. BPTI has a relatively broad specificity, inhibiting trypsin- as well as chymotrypsin- and elastase-like serine (pro)enzymes endowed with very different primary specificity. BPTI reacts rapidly with serine proteases to form stable complexes, but the enzyme:inhibitor complex formation may involve several intermediates corresponding to discrete reaction steps. Moreover, BPTI inhibits the nitric oxide synthase type-I and -II action and impairs K⁺ transport by Ca²⁺-activated K⁺ channels. Clinically, the use of BPTI in selected surgical interventions, such as cardiopulmonary surgery and orthotopic liver transplantation, is advised, as it significantly reduces hemorrhagic complications and thus blood-transfusion requirements. Here, the structural, inhibition, and bio-medical aspects of BPTI are reported.

Keywords: Aprotinin, Bovine basic pancreatic trypsin inhibitor, Kunitz inhibitor, Trypsin-kallikrein inhibitor, Structure, Inhibition properties, Bio-medical aspects.

****This paper is dedicated to Prof. Eraldo Antonini and Prof. Mario Guarneri who guided us in the early stages of our studies in the field of serine protease inhibition***

INTRODUCTION

The pancreatic Kunitz inhibitor, also known as aprotinin, bovine basic pancreatic trypsin inhibitor (BPTI), and trypsin-kallikrein inhibitor, is one of the most extensively studied globular proteins. It has proved to be a particularly attractive and powerful tool for studying protein conformation, folding, and dynamics as well as molecular bases of protein/protein interaction(s) and (macro)molecular recognition. BPTI has been a landmark protein in the early years of structural biology, lending itself to extensive crystallographic investigations as well as to the development of macromolecular NMR techniques. BPTI has a relatively broad specificity, inhibiting trypsin- as well as chymotrypsin- and elastase-like serine (pro)enzymes. Moreover, BPTI inhibits nitric oxide biosynthesis and impairs K⁺ transport by Ca²⁺-activated K⁺ channels. Since BPTI has a very high specificity for plasmin, its

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pharmacological use has been demonstrated to be particularly advisable in any clinical setting associated with hyperfibrinolysis, such as cardiopulmonary surgery and orthotopic liver transplantation. In these conditions, the increased generation of plasmin may cause severe haemorrhagic complications, whose incidence is reduced by BPTI, which limits blood-transfusion requirements (see [1-26]).

Kraut and coworkers independently discovered BPTI as a kallikrein inactivator in bovine lymph nodes in 1930 [27] and as a trypsin inhibitor in bovine pancreas by Kunitz and Northrop in 1936 [28]. BPTI has now been found in a variety of bovine organs and tissues, especially in those rich in mast cells (see [2,11,29-35]). Low molecular weight proteins or peptide fragments related to BPTI have been also identified in bovine organs and tissues (see [11,34,36-44]). BPTI-like inhibitors have been found not only in mammals, but also in other vertebrates and in invertebrates (see [11,34,45-60]).

Here, the structural, inhibition, and bio-medical aspects of BPTI are discussed.

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STRUCTURAL ASPECTS

The BPTI Primary Structure

The primary structure of BPTI (Fig. 1) was established by Kassel and Laskowski in 1965 [61] and by Anderer and Hornle in 1966 [62]. The single-chain polypeptide consists of 58 amino acid residues, and has a calculated M_r of 6512. Most of the variant amino acid residues can be explained by single-base mutations in the third position of the respective codons. The conservation of the Lys/Arg15(P_1)¹ and Cys residues (at positions 5, 14, 30, 38, 51, and 55) hold true for many BPTI-like inhibitors (Fig. 1) (see [2,11,31,42,43,45, 48-52,54,55,57-60]).

The BPTI Tertiary Structure

The crystal structure of monomeric BPTI was solved in 1970, at 2.5 Å resolution; different crystal forms have been isolated and studied by both X-ray crystallography and neutron diffraction (see [1,63-69]). Moreover, the crystal structures of BPTI mutants have been solved (see [69-76]). The structure of wild type and mutant monomeric BPTI has also been thoroughly characterized through nuclear magnetic resonance (see [4-6,24-26,69,77,78]). The structure of decameric BPTI has been also determined by X-ray crystallography (see [69,79,80]).

Monomeric BPTI shows a very compact pear-shaped structure (see Fig. 2, top panel). The reactive site bond in BPTI is that between Lys15(P_1) and Ala16(P_1') at the narrow end of the molecule. The relatively small hydrophobic core of BPTI consists of the side chains of the following residues: Phe4, Cys5-Cys55, Phe22, Tyr23, Cys30-Cys51, Phe33, Tyr35, and Phe45. BPTI-Asn43 is also an internal residue; three hydrogen bonding interactions from its side chain to main chain atoms participate in the overall molecular stability. There are two helical regions of secondary structure in BPTI. One and one-half turns (Pro2 to Glu7) of 3_{10} helix are near the *N*-terminus. Almost three turns of α -helix (Ser47 to Gly56) are found near the *C*-terminus. The side chain oxygen atoms of Ser47 and Glu49 form hydrogen bonds to amide nitrogen atoms at the start of the α -helix. There is a long twisted (by approximately 180°) double-stranded antiparallel b-loop from Ile18 to Tyr35. Antiparallel b-sheet hydrogen bonding is also made between residues Phe45 and Tyr21 and has prompted some to include this as a third strand of the antiparallel sheet. In the reactive site loop region there is a type I b-turn from Asn24 to Ala27. In this unusual turn region the carbonyl-oxygen atom of Asn24 receives two hydrogen bonds from the amide nitrogen atoms of Ala27 and Gly28. At the other end of these strands, Gly36 makes two hydrogen bonding interactions, Gly36 NH atom to Thr11 CO atom and Gly36 CO atom to the reactive site bond Ala16 NH atom (see [1,4,5,6,12,24-26,63-77]).

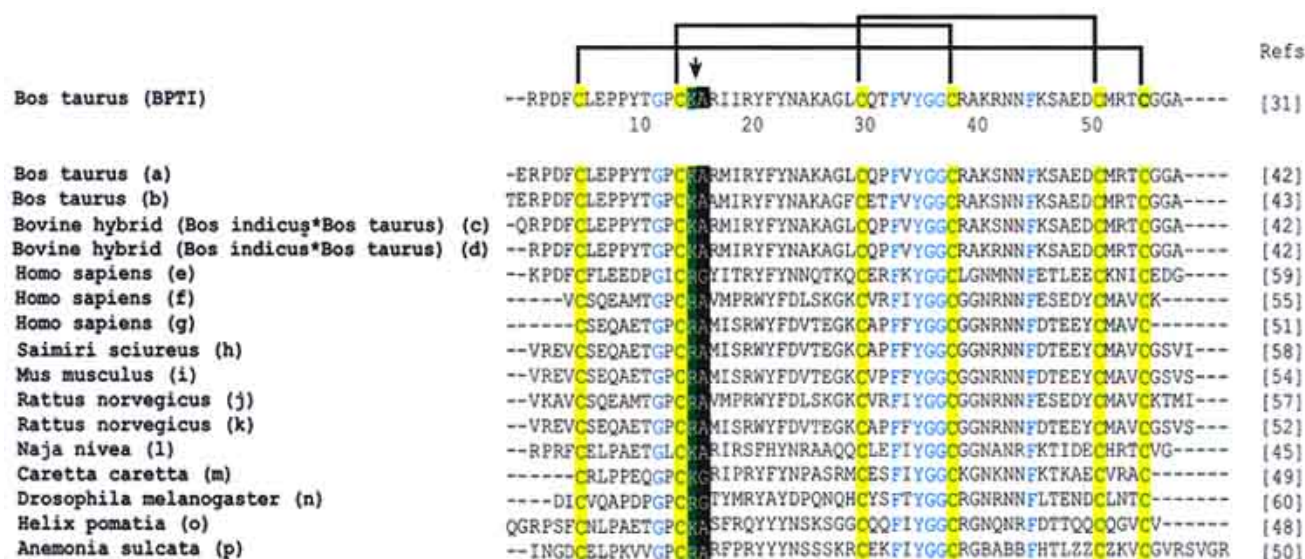


Fig. (1). Amino acid sequence alignment of BPTI (ID code: P00974) [31] and BPTI-like proteins: aprotinin analog (a; ID code: 1405218B) [42], serum basic protease inhibitor (b; ID code: P00975) [43], isoaprotinin G1 (c; ID code: S00371) [42], isoaprotinin G2 (d; ID code: S10063) [42], second Kunitz domain of tissue factor pathway inhibitor-b (e; ID code: AAD01700) [59], amyloid-b (A4) precursor-like protein 2 (f; ID code: NP_001633) [55], amyloid-b peptide (g; ID code: NP_000475) [51], Alzheimer's disease amyloid A4 (h; ID code: Q95241) [58], Alzheimer's disease amyloid A4 amyloidogenic glycoprotein (i; ID code: P12023) [54], amyloid-like protein 2 precursor (j; ID code: P15943) [57], Alzheimer's disease amyloid A4 protein homolog precursor (k; ID code: P08592) [52], venom basic protease inhibitor II (l; ID code: P00986) [45], chelonianin (m; ID code: P00993) [49], fat-spondin-P1 (n; ID code: NP_477435) [60], isoINHIBITOR K (o; ID code: P00994) [48], and snake-locks sea anemone (p; ID code: B27222) [50]. P_1 and P_1' residues forming the potentially scissile reactive site bond are highlighted in green and black respectively. The arrow indicates the P_1 - P_1' potentially scissile reactive site bond of BPTI. Amino acid numbering of BPTI is shown. Residues conserved in all proteins are in blue. Cysteine residues are highlighted in yellow. Disulphide bridges of BPTI are shown by brackets. The alignment was obtained with CLUSTAL W [192].

¹Serine (pro)enzyme residues have been numbered by taking as a reference the b. chymotrypsinogen A polypeptide chain [190]. The specificity subsites surrounding the catalytic center of serine (pro)enzymes have been identified as $S_n \dots S_1, S_1' \dots S_n'$. The amino acid residues forming the reactive site of BPTI have been labeled as $P_n \dots P_1, P_1' \dots P_n'$. P_1 - P_1' is the potentially scissile reactive site peptide bond of BPTI [191].

Decameric BPTI (see Fig. 2, bottom panel) resembles a compact globular particle where ten BPTI subunits are organized in the uncommon 52 point group symmetry (*i.e.*, orthogonal 5-fold and 2-fold axes). Interestingly, the reactive sites of BPTI subunits are facing the central pore of the decamer (see [69,79,80]).

The Serine (Pro)enzyme:BPTI Complexes

The crystal structure of wild type and mutant BPTI was analyzed in molecular complexes with wild type and mutant *b.* and *r.* b-trypsin, *b.* anhydrotypsin, *b.* and *r.* trypsinogen, *b.* trypsinogen:Ile-Val and :Val-Val adducts, *p. p.* kallikrein-A, wild type and recombinant *h.* a-thrombin, *h.* extracellular tissue factor bound to the coagulation factor VIIa, *h.* matriptase SP-1, and *b.* a-chymotrypsin (see [16-17,64,69, 81-102]).

The serine (pro)enzyme:BPTI complexes display the conventional mushroom shape; the serine (pro)enzyme represents the head of the mushroom and BPTI is the stalk. Unusually, the *b.* a-chymotrypsin:BPTI complex adopts a compact dimeric structure (two *b.* a-chymotrypsin:BPTI complex moieties related by a non-crystallographic two-fold

axis), the two BPTI molecules being roughly antiparallel to each other, each of them simultaneously contacting both *b.* a-chymotrypsin molecules (see Fig. 3). On the other hand, the *h.* matriptase:BPTI complex adopts a loose dimeric assembly in the crystalline state; the two mushroom shaped complex moieties are aired by their heads (see [16-17,64,69,81-102]).

The main contact region between serine (pro)enzymes and BPTI occurs at the inhibitor reactive site loop, between residues Pro13(P₃) and Arg17(P₂'). In this region, the BPTI reactive site forms a sort of antiparallel β -structure with the active site region of the serine (pro)enzyme, establishing direct N and O polypeptide backbone hydrogen bonds. From comparison of the three-dimensional models, it can be easily appreciated that BPTI binding to the serine (pro)enzymes has very little effect on its structure. The intermolecular hydrogen bonding occurring at sites P₃-P₂', around the (re)active site, are essentially maintained in the different binary complexes, reflecting both conservation of active site organization in the serine (pro)enzymes, as well as the rigidity of the reactive site loop in the BPTI molecule (see [16-17,64,69,81-102]).

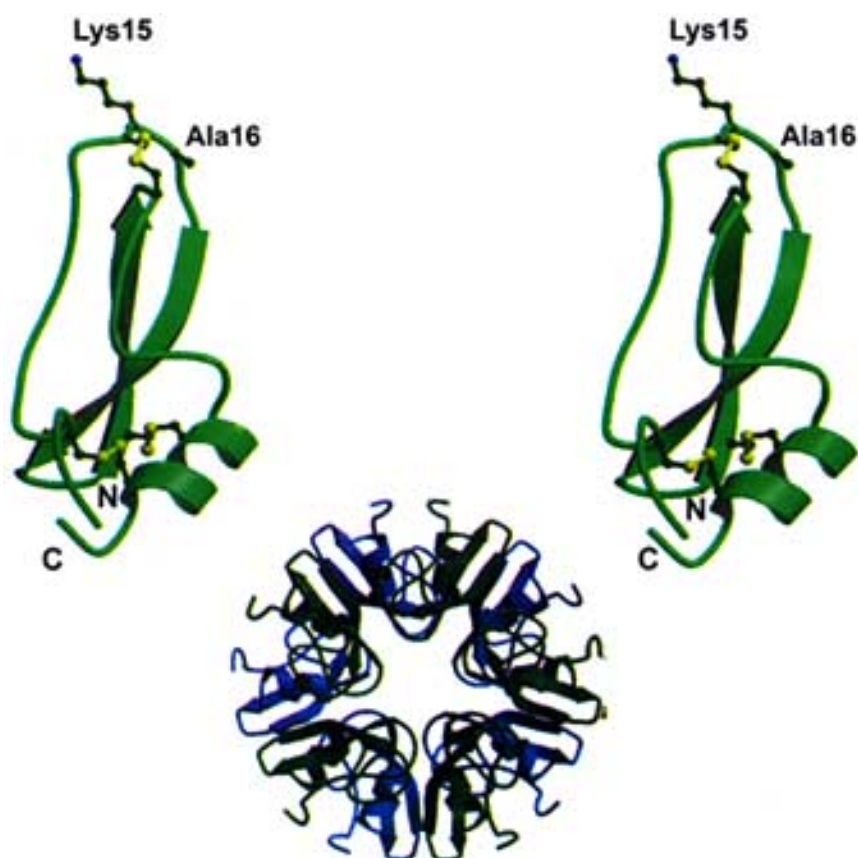


Fig. (2). *Top panel.* Three-dimensional structure of monomeric BPTI (PDB code: 4PTI) [64]. Lys15(P₁) and Ala16(P₁') residues forming the potentially scissile reactive site bond are shown. Disulphide bridges are shown in yellow. *Bottom panel.* Three-dimensional structure of decameric BPTI (PDB code: 1BHC) [79]. BPTI molecules are colored in green and blue. The reactive sites of BPTI molecules are facing the central pore of the decamer. The picture has been generated with MOLMOL [193].

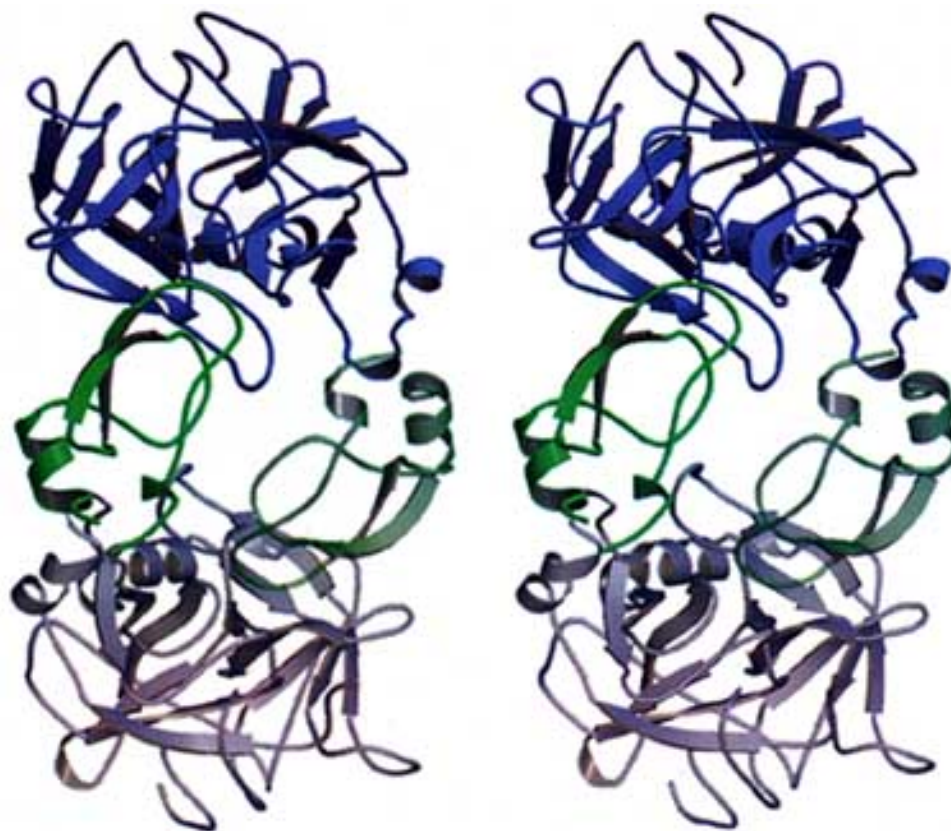


Fig. (3). The mushroom shape of the b. a-chymotrypsin:BPTI complex adopting a compact dimeric structure (PDB code: 1MTN) [94]. B. a-chymotrypsin molecules are drawn in dark and light blue and BPTI molecules in dark and light green. The picture has been generated with MOLMOL [193].

The serine (pro)enzyme:BPTI complexes are generally taken as stable species resembling intermediates in the normal (pro)enzyme substrate-cleavage reaction. The BPTI reactive site bond Lys15(P₁)-Ala16(P₁') is properly located in the serine (pro)enzyme active site such that the carbonyl carbon atom of the scissile peptide bond is close to the nucleophile Ser195 OG atom. Moreover, the carbonyl O atom of BPTI-Lys15(P₁) residue is stabilized by two hydrogen bonds with the N amide atoms of the serine (pro)enzyme residues Gly193 and Ser195, *i.e.* the so-called 'oxyanion binding site'. Interestingly, the geometry of the BPTI-Lys15(P₁) reactive site carbonyl is grossly intermediate between that of a planar peptide and that of a tetrahedral intermediate (see [16-17,64,69,81-102]).

The positively charged BPTI-Lys15(P₁) residue fills the S₁ primary specificity subsite of trypsin(ogen)-like serine (pro)enzymes, forming polar interactions with the invariant negatively charged Asp189 residue (see Fig. 4, top panel). On the contrary, the positively charged BPTI-Lys15(P₁) residue does not fill the uncharged S₁ subsite of b. a-chymotrypsin, but, rather, finds a hydrogen bonding compensation in the carbonyl O atoms of the enzyme surface residues Gly216 and Ser217 (see Fig. 4, bottom panel). Thus, at variance with the trypsin(ogen)-like serine (pro)enzyme:BPTI complexes, the BPTI-Lys15(P₁) side

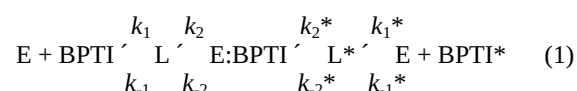
chain adopts a rather bent conformation in the b. a-chymotrypsin:BPTI complex (see [16-17,64,69,81-102]).

Binding of BPTI to the b. trypsinogen active center and of activating dipeptides (*e.g.*, Ile-Val and Val-Val), mimicking the *N*-terminal residues of b. b-trypsin, to the Ile-Val binding pocket induces the active conformation of the zymogen. Indeed, the so-called activation domain and the (pro)enzyme active site region are functionally coupled. Both BPTI and activating dipeptides induce rigidity of the flexible polypeptide loops Ile16-Gly19, Gly142-Pro152, Gly184-Gly193, and Gly216-Asn223, building up the serine (pro)enzyme activation domain (see [9,15-17,64,69,81-105]).

INHIBITION PROPERTIES

The Minimum Mechanism for BPTI Binding to Monomeric Serine (Pro)enzymes

The generally accepted minimum reaction scheme for BPTI binding to monomeric (chymo)trypsin(ogen)-like serine (pro)enzymes is:



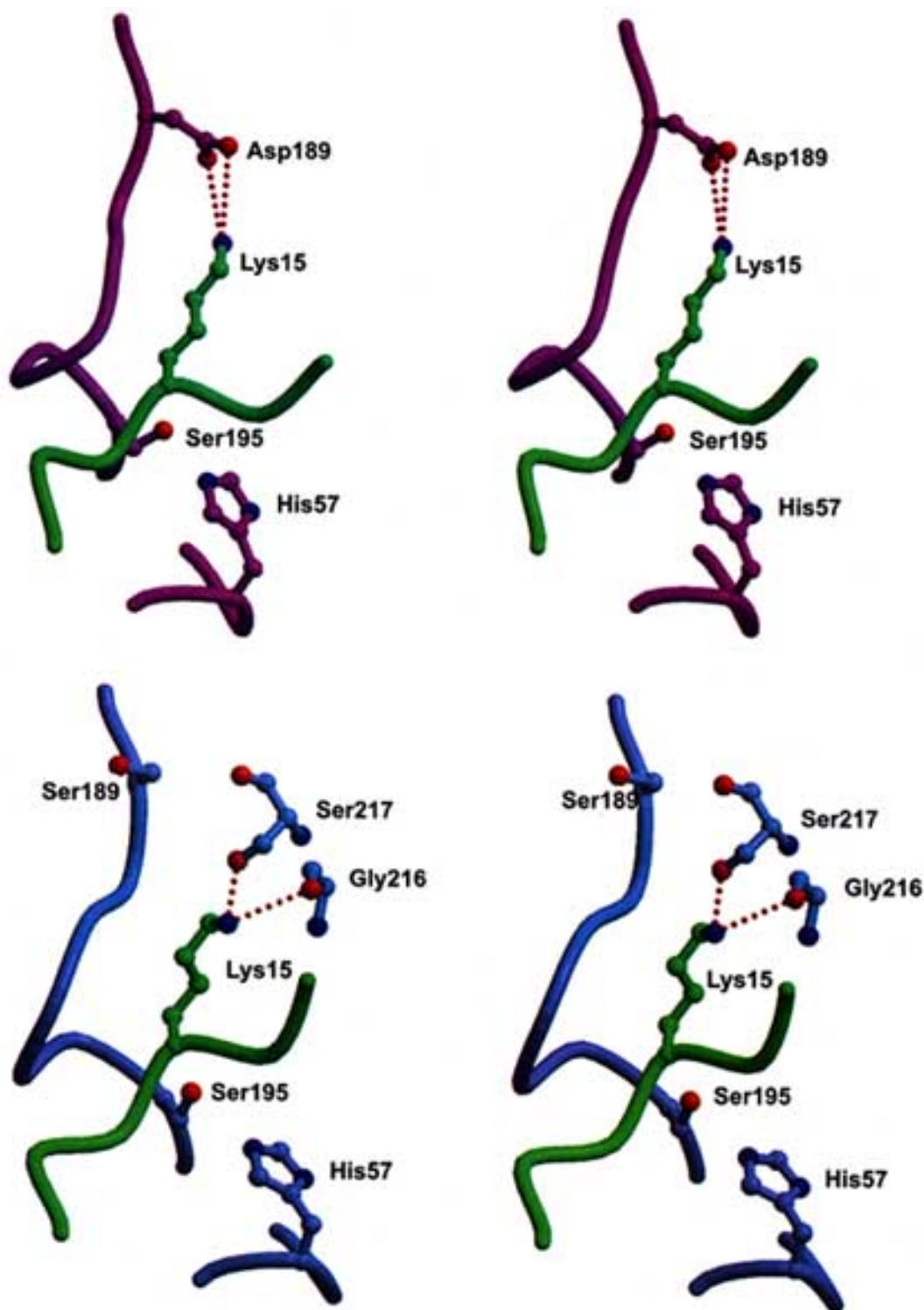


Fig. (4). *Top panel.* Binding mode of the BPTI-Lys15(P₁) reactive site residue to the b. b-trypsin primary specificity subsite S₁ (PDB code: 2PTC) [64]. The b. b-trypsin molecule is drawn in magenta and the BPTI molecule in green. The polar interactions between the BPTI-Lys15(P₁) reactive site residue and the Asp189 residue of b. b-trypsin are shown as red dotted lines. *Bottom panel.* Binding mode of the BPTI-Lys15(P₁) reactive site residue to the b. a-chymotrypsin active center (PDB code: 1MTN) [94]. The b. a-chymotrypsin molecule is drawn in blue and the BPTI molecule in green. The side chain of the BPTI-Lys15(P₁) residue is located outside the primary specificity subsite S₁ of b. a-chymotrypsin. The Ne atom of the BPTI-Lys15(P₁) residue forms two hydrogen bonds with the carbonyl oxygen atom of the Gly216 residue and the Ob atom of the Gly217 residue (red dotted lines). The picture has been generated with MOLMOL [193].

where E is the serine pro(enzyme), BPTI is the virgin inhibitor, BPTI* is the modified inhibitor with the reactive site bond Lys15(P₁)-Ala16(P₁') split, L and L* are loose complexes of E with BPTI and BPTI*, respectively, E:BPTI is the final serine (pro)enzyme:BPTI complex, k_1 is the second order rate constant for the formation of the loose complex L, k_{-1} is the first order rate constant for the dissociation of the loose complex L, k_2 is the first order rate constant for the conversion of the loose complex L to the final E:BPTI complex and represents the rate limiting step for E:BPTI complex formation, k_{-2} is the first order rate constant for the conversion of the E:BPTI complex to the loose complex L and represents the rate limiting step for E:BPTI complex dissociation, $K (= k_2/(k_{-2}(k_{-1}/k_1)))$ is the apparent dissociation equilibrium constant for E:BPTI complex formation starting from E and BPTI, $K_1 (= k_{-1}/k_1)$ is the dissociation pre-equilibrium constant for L formation starting from E and BPTI, $K_2 (= k_2/k_{-2})$ is the isomerization equilibrium constant for E:BPTI formation starting from L, k_1^* is the second order rate constant for the formation of the loose complex L*, k_{-1}^* is the first order rate constant for the dissociation of the loose complex L*, k_2^* is the first order rate constant for the conversion of the loose complex L* to the final E:BPTI complex and represents the rate limiting step for E:BPTI complex formation, k_{-2}^* is the first order rate constant for the conversion of the final E:BPTI complex to the loose complex L* and represents the rate limiting step for E:BPTI complex dissociation, $K^* (= k_{-2}^*/(k_2^*(k_{-1}^*/k_1^*)))$ is the apparent dissociation equilibrium constant for E:BPTI complex formation starting from E and BPTI*, $K_1^* (= k_{-1}^*/k_1^*)$ is the dissociation pre-equilibrium constant for L* formation starting from E and BPTI*, and $K_2^* (= k_2^*/k_{-2}^*)$ is the isomerization equilibrium constant for E:BPTI formation starting from L*. Under conditions where both BPTI and BPTI* coexist in the reaction mixture, the overall equilibrium constant is $K_{obs} = K/(1+K_{hyd})$; $K_{hyd} (= [BPTI^*]/[BPTI])$ represents the equilibrium constant for the hydrolysis of the inhibitor. For all the serine (pro)enzyme/BPTI and /BPTI* systems investigated, the concentration dependence of the rate constant for E:BPTI complex formation is second order at low inhibitor concentrations and tends to first order at high BPTI or BPTI* concentrations. Interestingly, b. α -chymotrypsin and b. β -trypsin are inhibited by BPTI* and catalyze the resynthesis of the reactive site peptide bond Lys15(P₁)-Ala16(P₁

BPTI Binding to Monomeric Serine (Pro)enzymes: Thermodynamic Aspects

BPTI has a relatively broad specificity, inhibiting trypsin- as well as chymotrypsin- and elastase-like serine (pro)enzymes endowed with very different primary specificity. The affinity of BPTI for serine proteases endowed with different primary specificity (e.g., trypsin, chymotrypsin, and elastase) is substantially affected by enzyme(S₁):inhibitor(P₁) contact(s) (see Tables 1 and 2 and Fig. 5) (see [100,138,139,142]). Thus, the replacement of the BPTI-Lys15(P₁) residue with an apolar side chain, as *nor*Val, *nor*Leu or Phe, forces the inhibitor P₁ hydrophobic

side chain in the protease S₁ specificity pocket, which hosts Asp189, causing a six to nine orders of magnitude drop in the inhibitor affinity [114,132]. On the other hand, the affinity of wild-type BPTI for b. α -chymotrypsin is quite close to that of the semisynthetic BPTI variants bearing the apolar P₁ residues *nor*Val, *nor*Leu or Phe [114,132]. Such an observation has been interpreted accounting that the cost of restructuring the BPTI-Lys15(P₁) side chain outside the b. α -chymotrypsin S₁ subsite is essentially compensated by hydrogen bonding to Gly216 O and Ser217 O atoms (see Fig. 4, bottom panel) [94]. A semisynthetic approach to modulate the affinity of BPTI for h. l. elastase demonstrates the individual contribution of the P₁ side chain to the inhibition of a particular target protease; the BPTI-Val15(P₁) mutant is the most powerful inhibitor for h. l. elastase (see Table 2) [111,132].

BPTI variants mutated in the P₄, P₃, P₁, P₁', P₂', and P₃' positions of the reactive binding loop were used to probe the role of amino acid substitutions on binding to trypsin-, chymotrypsin-, and elastase-like serine proteases. Interestingly, mutations at positions P₃ and P₁ induce cumulative effects increasing the affinity of BPTI variants for trypsin-like enzymes (i.e., values of K decrease from 2.2- for h. α -plasmin inhibition to 4,000-fold for h. factors X_a: and XII_a:BPTI complex formation). On the other hand, mutations at the P₄ site always decrease the affinity BPTI variants for trypsin-like serine proteases (i.e., values of K for enzyme:BPTI complex formation increase up to 1,000-fold) (see [139-141,143-145]).

The affinity of BPTI for serine proteases endowed with Lys/Arg substrate primary specificity (e.g., kallikrein, thrombin, factor Xa, plasmin, and urokinase) can be substantially affected by structural factors outside the catalytic center region (see Table 1) [14]. Thus, an affinity decrease of four to ten orders of magnitude can be ascribed to the influence of protease loops, which limit the inhibitor accessibility (e.g., external loops in p. p. kallikrein, and b- and g-loops in h. α -thrombin). Removal of such structural constraints increases protease:BPTI affinity. Moreover, the removal of the Pro60B, Pro60C, and Trp60D residues from the 60-insertion loop and the Glu192 $\rightarrow$ Gln mutation enhance h. α -thrombin inhibition by BPTI. The Pro60B-Pro60C-Trp60D deletion and the Glu192 $\rightarrow$ Gln mutation contribute comparably and additively to the binding energy of h. α -thrombin to BPTI. H. α -thrombin having both the Pro60B-Pro60C-Trp60D deletion and the Glu192 $\rightarrow$ Gln mutation binds BPTI with $K = 3.5 \times 10^{-12}$ M; the value of K for wild-type BPTI binding to h. α -thrombin is 8.0×10^{-4} M (see Table 1) (see [122,125,129,130,135,136]).

The low affinity of BPTI for b. trypsinogen has been attributed to the loss of free energy required to induce the b. β -trypsin-like active conformation in the proenzyme (i.e., to achieve structure and rigidity in the polypeptide loops building up the serine (pro)enzyme activation domain). Binding of the activating peptides (e.g., Ile-Va1), mimicking the activating N-terminus of activated b. β -trypsin, to the Ile-Val pocket of b. trypsinogen increases the affinity of BPTI. Accordingly, the affinity of activating peptides for b. trypsinogen increases by four orders of magnitude in the

Table 1. Values of the Apparent Thermodynamic and Kinetic Parameters for the Binding of Wild-Type BPTI to Representative Serine (Pro)enzymes.^a

(Pro)enzyme	<i>K</i> (M)	<i>k</i> ₁ (M ⁻¹ s ⁻¹)	<i>k</i> ₋₂ (s ⁻¹)	References
H. trypsin-1	< 2×10 ⁻¹⁰	1.0×10 ⁵	< 2×10 ⁻⁵	[137]
H. trypsin-2	< 10 ⁻⁹	2.4×10 ⁵	< 2×10 ⁻⁴	[137]
B. b-trypsin	6.0×10 ⁻¹⁴	1.1×10 ⁶	6.6×10 ⁻⁸	[106]
B. pseudotrypsin	9.0×10 ⁻⁹	7.0×10 ⁴	6.3×10 ⁻⁴	[106]
B. anhydrotrypsin	1.1×10 ⁻¹³	7.7×10 ⁵	8.5×10 ⁻⁸	[120]
B. trypsinogen	2.0×10 ⁻⁶	2.0×10 ⁴	4.0×10 ⁻²	[91,104,105,108,122; Ascenzi, P., unpublished results]
B. trypsinogen:Ile-Val	<1×10 ⁻⁹	2.0×10 ⁵	<2×10 ⁻⁴	[104,105,122; Ascenzi, P., unpublished results]
H. u. kallikrein	9.1×10 ⁻¹¹	---	---	[122,130]
P. p. b-kallikrein-A	8.4×10 ⁻¹⁰	3.0×10 ⁵	2.7×10 ⁻⁴	[122,130; Ascenzi, P., unpublished results]
P. p. b-kallikrein-B	8.4×10 ⁻¹⁰	3.0×10 ⁵	2.7×10 ⁻⁴	[122,130; Ascenzi, P., unpublished results]
H. a-thrombin	8.0×10 ⁻⁴	4.0×10 ⁴	7.0×10 ¹	[125,129]
H. b-thrombin	4.0×10 ⁻⁴	5.0×10 ⁴	3.0×10 ¹	[125,129]
H. g-thrombin	1.1×10 ⁻⁴	8.5×10 ⁴	1.0×10 ¹	[125,129]
H. des-Pro-Pro-Trp-thrombin	1.6×10 ⁻⁸	---	---	[135]
H. Glu192/ÆGln-thrombin	2.4×10 ⁻⁸	---	---	[136]
B. factor Xa	4.8×10 ⁻⁶	---	---	[126]
H. factor Xa	4.8×10 ⁻⁶	---	---	[126]
H. factor XII	9.1×10 ⁻⁵	---	---	[139]
H. protein C	1.6×10 ⁻⁷	---	---	[139]
H. a-plasmin	1.9×10 ⁻¹⁰	---	---	[139]
H. plasmin-Glu1	1.0×10 ⁻⁹	1.6×10 ⁵	1.6×10 ⁻⁴	[127]
H. plasmin-Lys77	8.3×10 ⁻¹⁰	1.8×10 ⁵	1.5×10 ⁻⁴	[127,131]
H. plasmin-Val442	7.7×10 ⁻¹⁰	2.0×10 ⁵	1.5×10 ⁻⁴	[127]
H. plasmin-Val561	6.3×10 ⁻⁹	1.1×10 ⁵	6.9×10 ⁻⁴	[127]
H. urokinase (<i>M_r</i> 33,000)	2.0×10 ⁻⁵	---	---	[128]
H. urokinase (<i>M_r</i> 54,000)	2.0×10 ⁻⁵	---	---	[128]
B. tryptase ^b	1 st site 2.6×10 ⁻⁸	---	---	[33]
	2 nd site 2.8×10 ⁻⁹	---	---	[33]
	3 rd site 4.4×10 ⁻⁷	---	---	[33]
	4 th site 4.4×10 ⁻⁵	---	---	[33]
H. chymotrypsin-A	1.7×10 ⁻⁷	~10 ⁴	~2×10 ⁻³	[137]
B. a-chymotrypsin	1.5×10 ⁻⁹	3.7×10 ⁵	5.5×10 ⁻⁴	[107,115,132,134]
H. cathepsin G	4.0×10 ⁻⁶	2.3×10 ⁵	9.0×10 ⁻¹	[133]
H. l. elastase	3.5×10 ⁻⁶	---	---	[132]
H. p. elastase	> 2×10 ⁻³	---	---	[137]
P. p. elastase	1×10 ⁻³	---	---	[141]

^aThe following notation applies: $K = k_{-2}/k_1$ (see Scheme 1).^bBPTI binding to homotetrameric b. tryptase is characterized by a complex functional interplay among the four inhibitor binding sites (see Scheme 2 and [33]).

Table 2. Values of Apparent K for the Binding of Semisynthetic BPTI Mutated at the P_1 Position to h. l. Elastase

P_1 residue	P_1 side chain	K (M)
Lys (wild-type BPTI)	$-(CH_2)_4NH_2$	3.5×10^{-6}
Gly	-H	7.0×10^{-9}
Ala	$-CH_3$	2.5×10^{-9}
2-Aminobutyric acid	$-CH_2CH_3$	2.3×10^{-10}
<i>nor</i> Val	$-(CH_2)_2CH_3$	1.6×10^{-10}
Val	$-CH(CH_3)_2$	1.1×10^{-10}
<i>nor</i> Leu	$-(CH_2)_3CH_3$	2.7×10^{-9}
Leu	$-CH_2CH(CH_3)_2$	2.9×10^{-9}
Ile	$-CHCH_3CH_2CH_3$	4.1×10^{-10}
<i>tert</i> Leu	$-C(CH_3)_3$	1.5×10^{-8}
<i>tert</i> butylAla	$-CH_2C(CH_3)_3$	6.7×10^{-8}

From [132].

presence of macromolecular inhibitors (*e.g.*, BPTI) (see Table 1) (see [104,105,122]).

The serine (pro)enzyme:BPTI complex formation is an endothermic and entropy-driven process, it is characterized by an enthalpy-entropy compensation effect. The positive DS° term should reflect the expulsion of the serine (pro)enzyme-bound water molecules upon BPTI binding (see [107,125-131,133,146]).

Interestingly, BPTI binding to serine (pro)enzymes may be allosterically controlled. In particular, calcium and lanthanide ions (*e.g.*, terbium) affect BPTI binding to b. α -chymotrypsin. Indeed, the cation binding induces conformational transitions on b. α -chymotrypsin that destabilize the E:BPTI complex formation affecting the interaction of BPTI with the secondary specificity subsites of the serine protease. The mutual correlation between BPTI binding to b. α -chymotrypsin and calcium release from the serine protease (and *vice versa*) could assume an important physiological significance linking parameters, such as calcium concentration and the activity levels of proteolytic enzymes, which are both of great importance for the cell life (see [134]).

BPTI Binding to Monomeric Serine (Pro)enzymes: Kinetic Aspects

BPTI reacts rapidly with serine proteases to form stable complexes, but the E:BPTI complex formation may involve several intermediates corresponding to discrete reaction steps. In particular, BPTI binding to h. trypsin-1 and -2, b. pseudotrypsin, b. anhydrotrypsin, b. trypsinogen(:Ile-Val), p. p. b-kallikrein-A, p. p. b-kallikrein-B, h. α -, b-, and g-thrombin, h. plasmin-Glu1, -Lys77, -Val442, and -Val561, h. chymotrypsin-A, b. α -chymotrypsin, and h. cathepsin G as well as BPTI* binding b. α -chymotrypsin may be described within Scheme 1; it applies also to wild-type and mutant

BPTI binding to Ca^{2+} -activated K^+ channels. On the other hand, a multistep process may account for BPTI and BPTI* binding to b. b-trypsin (see [3,7,11,22,91,106-108,115-120,123-125,129,131,133,137]).

The formation of the b. b-trypsin:BPTI complex was monitored, making use of three different signals: proflavine displacement, optical density changes in the ultraviolet region, and the loss of the catalytic activity. The rates of the reactions indicated by the three different signals are similar at neutral pH, but diverge at low pH. At pH 3.5, proflavine displacement precedes the optical density changes in the ultraviolet and the loss of enzyme activity by several orders of magnitude in time (see Fig. 6) [123]. These data indicate that the b. b-trypsin:BPTI complex formation is a multistage process and led to the prediction that BPTI addition to the b. b-trypsin:proflavine complex would remove proflavine inhibition and the enzyme would recover transiently its catalytic activity before being irreversibly inhibited by completion of BPTI binding. The kinetic evidences shown in (Fig. 7) verified this prediction, indicating that during the b. b-trypsin:BPTI complex formation one transient intermediate occurs, which is not able to bind proflavine but may bind and hydrolyze the substrate [124]. This peculiar catalytic behavior, is in line with the proposed reaction mechanism for the b. b-trypsin:BPTI complex formation, which postulates a sequence of distinct polar and apolar interactions at the contact area [123,124].

Values of the second order rate constant for BPTI binding to serine (pro)enzymes (*i.e.*, k_1) agree well with one another, ranging between $10^4 \text{ M}^{-1} \text{ s}^{-1}$ and $10^6 \text{ M}^{-1} \text{ s}^{-1}$ (see Table 1); this suggests that energetically equivalent limiting step(s) controls the binding process. On the other hand, values of k_2 for the serine protease/BPTI systems investigated widely diverge from $6.6 \times 10^{-8} \text{ s}^{-1}$, for the dissociation of the b. b-trypsin:BPTI adduct, to $7.0 \times 10^1 \text{ s}^{-1}$, for the destabilization of the h. α -thrombin:BPTI complex (see Table 1). Thus, differences in k_2 values for the serine

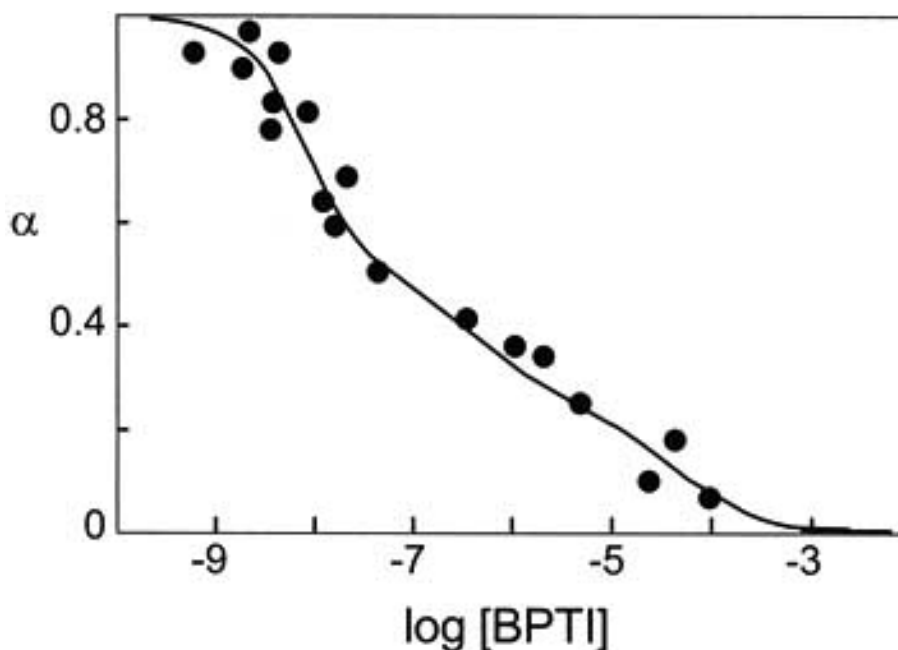


Fig. (5). Dependence of the fraction of the residual activity of b. trypase (α) on BPTI concentration (M). The continuous line was calculated according to the following equation:

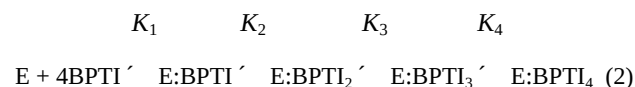
$$\alpha = \frac{1}{2} \cdot \frac{K_1[\text{BPTI}] + K_2[\text{BPTI}] + 2K_1K_2[\text{BPTI}]^2}{2 \cdot (1 + K_1[\text{BPTI}] + K_2[\text{BPTI}] + K_1K_2[\text{BPTI}]^2)} + \frac{1}{4} \cdot \frac{K_3[\text{BPTI}]}{1 + K_3[\text{BPTI}]} + \frac{1}{4} \cdot \frac{K_4[\text{BPTI}]}{1 + K_4[\text{BPTI}]}$$

with values of K given in Table 1. BPTI binding to b. trypase displays a complex interplay between positive and negative interaction effects. Modified from [33].

protease:BPTI complex destabilization mostly account for the affinity changes of the inhibitor binding, expressed by values K (see Table 1). On the other hand, differences in values of the second order rate constant for the binding of BPTI mutants to Ca^{2+} -activated K^+ channels mostly account for the inhibitor affinity (see [3,7,11,22,91,106-108,115-120,123-125,129,131,133,137]).

BPTI Binding to Tetrameric Trypsin

Trypsins are trypsin-like serine proteases found in the granules of mast cells where BPTI is also localized (see [147]). H. trypase is a ring-like tetramer formed by four identical subunits with the active sites facing the central pore [69,147-149]. The minimum reaction scheme for BPTI binding to tetrameric trypase is:



where K_1 , K_2 , K_3 , and K_4 are the apparent dissociation equilibrium constants for mono-, di-, tri-, and tetra-ligation of b. trypase by BPTI. A complex functional interplay between the four enzyme active centers has been envisaged for BPTI binding to tetrameric b. trypase. In particular, a weak positive cooperativity in inhibitor binding occurs for the formation of the trimeric complex composed of the

tetrameric b. trypase functional unit and two BPTI molecules. The presence of two inhibitor molecules strongly impairs the binding of the third BPTI molecule, the affinity of the fourth BPTI molecule to b. trypase being further reduced. The low affinity of BPTI for the third and the fourth binding site of b. trypase is likely due to steric hindrance exerted by the first and second BPTI molecules bound to b. trypase and facing the central pore of the tetrameric serine protease (see Fig. 5 and Table 1) [33].

Recombinant BPTI variants inhibit trypase TL2, a serine esterase from human T4+ lymphocytes, and HIV-1 replication in H9 cells. Interestingly, h. trypase TL2 is a binding component to HIV-1 glycoprotein gp120, playing a role in the HIV-1 infection mechanism. In contrast to wild type BPTI, the recombinant variant [Lys15(P₁)/EArg, Arg17(P₂')/EPhe, Met52/EGlu]-BPTI, with a reactive-site sequence homologous to the V3 loop of HIV-1 gp120, inhibits specifically h. trypase TL2 activity. On the other hand, the [Lys15(P₁)/ELeu, Arg17(P₂')/EPhe, Met52/EGlu]-BPTI variant is the most potent inhibitor of binding of HIV-1 gp120 to h. trypase TL2, significantly retarding syncytium formation [150].

BPTI Binding to Nitric Oxide Synthase

BPTI inhibits competitively NO synthesis catalyzed by rat nitric oxide synthase type I and II (r. NOS-I and NOS-II,

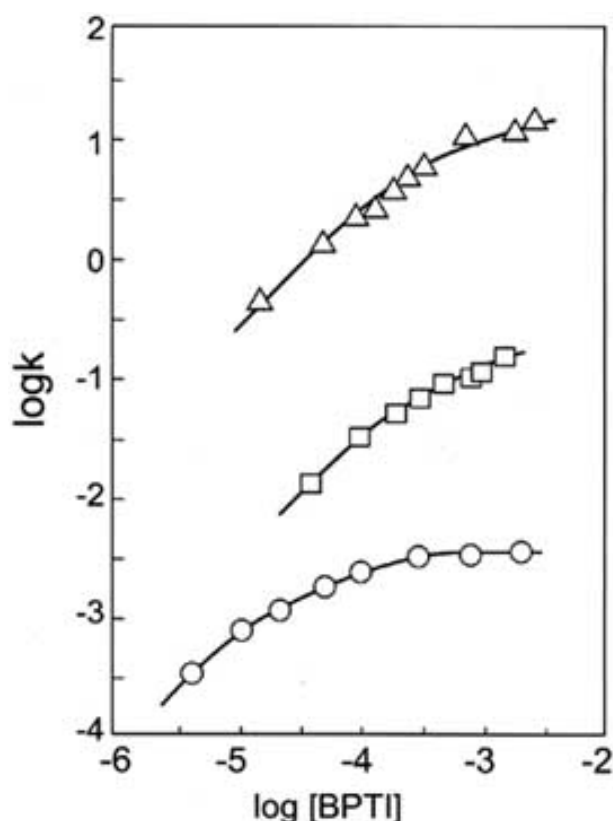


Fig. (6). Dependence of the apparent pseudo first order rate constant (k ; s^{-1}) on BPTI concentration (M) for the displacement of proflavine from b. b-trypsin (triangles), the spectral changes in the UV region of the serine protease (squares), and the loss of the enzymatic activity (circles). Continuous lines were calculated according to the following equation:

$$k = k_2 / (1 + K_1/[BPTI])$$

with the following parameters: proflavine displacement - $k_2 = 16.0 s^{-1}$ and $K_1 = 6.5 \times 10^{-4} M$, spectral changes in the UV region - $k_2 = 0.2 s^{-1}$ and $K_1 = 6.3 \times 10^{-4} M$, and loss of the enzymatic activity - $k_2 = 3.3 \times 10^{-3} s^{-1}$ and $K_1 = 4.0 \times 10^{-5} M$. The b. b-trypsin:BPTI complex formation is not an all-or-nothing process, but involves several intermediates corresponding to discrete reaction steps. Modified from [123].

respectively). BPTI binding to r. NOS-I and NOS-II conforms to a simple equilibrium; the r. NOS-I: and NOS-II:BPTI stoichiometry is 1:1. Values of K for BPTI binding to r. NOS-I and NOS-II are $5.0 \times 10^{-5} M$ and $7.8 \times 10^{-5} M$, respectively (see Table 1). As already reported for most r. NOS-I and NOS-II inhibitors, NO does not originate from BPTI. Although the r. NOS-I: and NOS-II:BPTI binding mode is unknown, it is worthwhile noting that BPTI displays six fully solvent exposed Arg residues at positions 1, 17, 20, 39, 42, and 53, which might be involved in the enzyme:inhibitor complex formation, as should be expected from r. NOS-I and NOS-II specificity properties (see [21]).

BPTI Binding to Ca^{2+} -activated K^+ Channels

Large conductance r. Ca^{2+} -activated K^+ channels contain a C-terminal intracellular binding site for BPTI. BPTI binding to Ca^{2+} -activated K^+ channels conforms to a simple equilibrium; the Ca^{2+} -activated K^+ channels:BPTI stoichiometry ranges from 1:1 to 1:4. Wild-type and mutant BPTI induce discrete substate events at the single-channel level. The dwell time of the substate exhibits relatively small changes (<9-fold) for the various mutants and is inversely related to the dissociation rate constant of wild-type and mutant BPTI (ranging between $0.64 s^{-1}$ and $8.8 s^{-1}$). On the other hand, the value of the second order rate constant for BPTI binding varies from $2.5 \times 10^4 M^{-1} s^{-1}$ to $1.2 \times 10^7 M^{-1} s^{-1}$ and exhibits a positive correlation with the net charge of BPTI, suggesting the presence of a negative electrostatic surface potential in the vicinity of the BPTI binding site. The substate current level is unaffected by most of the mutations except for substitutions of the BPTI-Lys15(P_i) residue. Different residues at this position modulate the apparent conductance of the BPTI-induced substate from 0% (Lys15(P_i)/EGly) to 55% (Lys15/Val) of the open state at +20 mV. This indicates that the BPTI inhibitory loop that contacts serine (pro)enzymes is involved in the formation of the Ca^{2+} -activated K^+ channel:BPTI complex. Binding of BPTI to one inhibitor-binding domain (out of four available) results in a conformation of the Ca^{2+} -activated K^+ channel that does not allow normal movement of the voltage sensor in that subunit. Thus, when one of the four subunits is in a conformation different from the others the channel opening is impaired. Such a BPTI-occupied channel rapidly fluctuates between the open and the closed state of the pore. Low-pass filtering of the rapidly flickering current gives rise to the appearance of a low-conductance substate in the single-channel record (see [22]).

BIO-MEDICAL ASPECTS

BPTI inhibits serine proteases involved in digestive and coagulation processes, such as trypsin, plasmin as well as tissue and plasma kallikrein. Hence over the past few decades the pharmacological applications of BPTI concerned clinical syndromes associated with dysfunctions of digestive and coagulation/fibrinolysis enzymes (see Table 3) (see [13,151-156]).

Clinical Syndromes Associated with Dysfunctions of Digestive Enzymes

BPTI has been used for more than three decades as an intensive-care drug for acute pancreatitis. The very high affinity of BPTI for trypsin makes theoretically BPTI an excellent drug able to antagonize the pathological effects of pancreatic serine proteases in acute pancreatitis. Human pancreas has been shown to synthesize a number of proteases including trypsin-1 and -2, chymotrypsin-A, and elastase [156-159]. BPTI forms very tight complexes with human pancreatic trypsin-1 and trypsin-2 (see Table 1) [160]. However, BPTI does not inhibit significantly human pancreatic chymotrypsin-A and human pancreatic elastase (see Table 1) [161]. The fact that BPTI is unable to inhibit

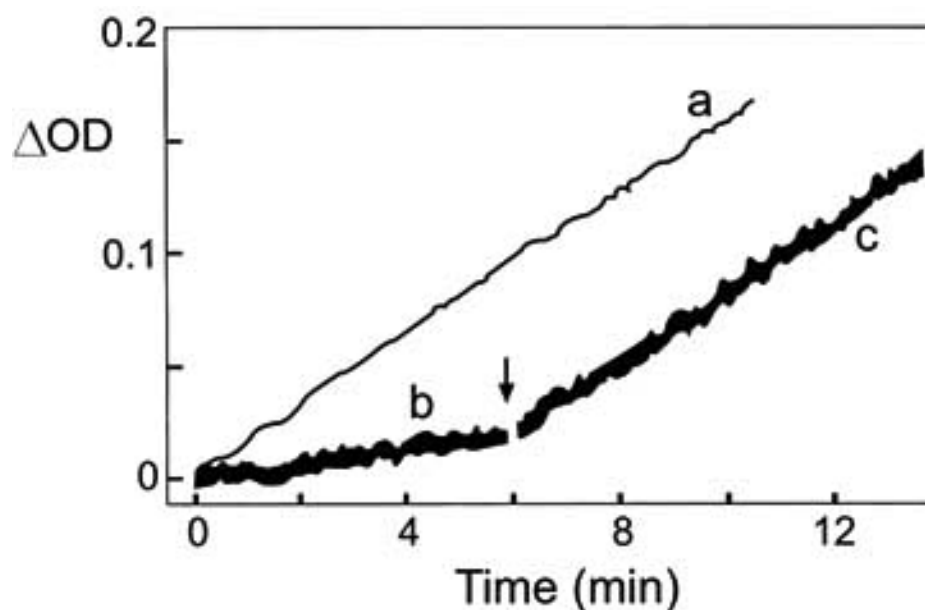


Fig. (7). B. b-trypsin inhibition by BPTI: a case for 'chronosteric effects'. *Trace a*, time course of the b. b-trypsin-catalyzed hydrolysis of Z-Lys-ONp in the absence of both proflavine and BPTI. *Trace b*, time course of the b. b-trypsin-catalyzed hydrolysis of Z-Lys-ONp in the presence of proflavine. *Trace c*, time course of the b. b-trypsin-catalyzed hydrolysis of Z-Lys-ONp in the presence of proflavine, but after the addition, indicated by the arrow, of BPTI. The transient activation stage of b. b-trypsin on addition of BPTI will be followed by the full inactivation of the serine protease due to the completion of the b. b-trypsin:BPTI complex formation. Modified from [124].

Table 3. Pharmaceutical and Clinical Aspects of BPTI

Pharmacodynamics
High-affinity competitive inhibition of plasmin and kallikrein
No effect on the biosynthesis of clotting and fibrinolytic factors
No significant effect on platelet functions
No significant pro-thrombotic effect
Very low toxicity and weak immunogenicity
Distribution sites
<i>Extracellular compartment:</i> rapid distribution ($t_{1/2} \sim 45$ min) into extracellular compartment after venous administration.
<i>Kidney:</i> 80-90% of an administered dose is temporarily stored in proximal renal tubular epithelium prior to elimination.
Pharmacokinetics
Renal metabolism, with $t_{1/2} \sim 20$ hours after loading intravenous injection
No organ deposition
Antithrombotic indications
Cardiac surgery
Orthotopic liver transplantation
Acute promyelocytic leukemia (unproved efficacy)
Major surgery in patients under antiaggregating agents (e.g., aspirin, thienopyridines)

the bulk of pancreatic proteases might explain why its therapeutic efficacy in digestive dysfunctions has sometimes been questioned [162]. Prospective, randomized trials treating human acute pancreatitis with antiproteolytic drugs (BPTI, gabexate mesylate, and even fresh frozen plasma)

have indeed failed to show any benefit in this clinical setting [162]. Thus, clinically, BPTI therapy has no significant effect on the course of severe acute pancreatitis [153].

Clinical Syndromes Associated with Dysfunctions of Coagulation/Fibrinolysis Enzymes

BPTI use in various dysfunctions of the hemostatic/fibrinolytic system shows a good efficacy in clinical trials. Activation of the coagulation cascade may occur through activation of some serine protease precursors, such as the coagulation factor XII [163]. The latter triggers the so called contact system, which is initiated by the binding of the plasma factor XII to a negatively charged surface. Upon this interaction, the plasma factor XII autoactivation occurs, converting it to factor XIIa. The presence of small amounts of factor XIIa convert pre-kallikrein to kallikrein [164], which in turn activates factor XII, cleaves high-molecular-weight kininogen, generating the vasoactive peptide bradykinin, and converts pro-urokinase to urokinase. Fibrinolysis consists of plasminogen activation, which, converted to the enzyme plasmin, ultimately digests fibrin fibrils. Plasminogen is activated by urokinase and by tissue plasminogen activator [163]. It is known that BPTI inhibits kallikrein, thus indirectly inhibiting the formation of activated factor XII [164]. Hence, BPTI inhibits the initiation of both coagulation and fibrinolysis, induced by the contact of blood with a foreign surface. By contrast, there is no evidence that BPTI could affect the platelet function [165-167], although the weak inhibition of the thrombin catalytic activity may affect in part the thrombin-induced platelet

activation, via competitive inhibition of the protease-activated receptor-1 hydrolysis [23].

The broad antiproteolytic action of BPTI suggested its use in reducing blood loss in patients undergoing cardiac surgery, which enhances not specific plasma proteolysis. In randomized studies it was shown that blood loss in patients undergoing repeated cardiac surgery, valve replacement and coronary-artery bypass grafting, and who were treated with BPTI was up to 80 percent less and transfusion requirements were up to 60 percent lower than in the control groups [168-175]. BPTI is also effective in operations characterized by particularly large blood loss, such as those in patients taking aspirin [169], in patients with endocarditis [176], and in patients undergoing heart transplantation [177].

A relatively low dose of BPTI in bolus (2 million kallikrein inactivation units, KIU; the inhibitory efficacy of BPTI is usually expressed in KIU) results in a reduction of approximately 35 percent in blood loss and a 50 percent reduction in transfusion requirements, as compared with no treatment, in a total of 20 patients undergoing orthotopic liver transplantation [178]. In a subsequent study of a larger dose (2 million KIU after the induction of anesthesia, followed by an infusion of 0.5 million KIU per hour during the operation) in a total of 24 patients, transfusion requirements in the treated patients were less than a third of those in the control patients [179]. These studies, however, were neither randomized nor controlled. It is worthy of note that in one small, randomized study of a total of 20 patients, BPTI was not more effective than placebo [180].

Acute promyelocytic leukemia is often associated with a severe hemostatic disorder, caused by the release of procoagulant and fibrinolytic substances from leukemic blasts. The coagulation profile may exhibit disseminated intravascular coagulation and fibrinolysis or proteolysis. Therefore, antifibrinolytic agents and particularly BPTI have been used to prevent severe bleedings. However, in a multicenter study on 268 consecutive patients with acute promyelocytic leukemia, neither the overall incidence of early hemorrhagic death nor the consumption of packed red blood cells were affected by BPTI treatment compared to other two groups treated with heparin or supportive therapy alone [151]. However, it has to be outlined that the outcome of this disorder has been dramatically ameliorated by the introduction of all-*trans*-retinoic acid, which induces a rapid correction of the hemostatic abnormalities [156].

Effect of BPTI on Inflammatory Phenomena

Recently, several studies have shown a beneficial effect of BPTI on inflammatory phenomena induced by extracorporeal circulation. Indeed, extracorporeal circulation used in major heart surgery induces a systemic inflammatory response, through increase of several cytokines, such as IL-6, which may adversely affect organ function. BPTI but not other antifibrinolytic agents, such as ϵ -aminocaproic acid, attenuate the rise in the proinflammatory cytokine IL-6 in patients under extracorporeal circulation [155]. Further studies are needed to validate the clinical benefit of BPTI

treatment on organ injury induced by an extracorporeal circulation-linked inflammatory response.

BPTI Dosage and Warnings

BPTI is usually given in bolus at the dose of 2-4 millions KIU intravenously, and then with continuous infusion of about 0.5 millions KIU/hour intravenously. Plasma concentration of about 125 KIU/ml is necessary to inhibit plasmin, while plasma concentration ranging between 300 and 500 KIU/ml inhibits kallikrein [181]. BPTI is a heterologous polypeptide, and therefore it can cause weak hypersensitivity reactions, particularly after repeated exposure within six months [182]. Interestingly, human anti-BPTI antibodies are mainly directed against the reactive site of BPTI [183].

BPTI has been hypothesized to cause venous and arterial thrombosis and thus the occlusion of coronary-bypass grafts and other vascular grafts. In controlled studies with coronary angiography, however, BPTI did not lead to an increased rate of early occlusion of saphenous-vein or internal-mammary-artery grafts [175,184-186]. Likewise, in prospective, placebo-controlled, randomized studies of BPTI-treated patients who had undergone coronary-artery bypass surgery, there was no increase in the occurrence of myocardial infarction or death [187]. Pooled analysis of data from studies of coronary-artery bypass surgery indicates a lower prevalence of stroke among the patients treated with BPTI than treated with placebo [188].

CONCLUSIONS

BPTI is one of the most extensively studied small globular proteins that has proved through the years as a particularly attractive and powerful tool for studying protein conformation, protein/protein interaction, and molecular recognition. BPTI binding to serine (pro)enzymes is not a single-event process. Rather, it consists of a space-time series of events involving molecular flexibility, induced fit, entropy, enthalpy, and free energy changes, solvent, as well as the achievement of specific hydrogen bonds, hydrophobic, van der Waals, and electrostatic interactions (see [9,11,12,15-17,19,89-91,104-105,122-124,129,135,136]).

The stereochemical analysis of the serine (pro)enzyme/BPTI systems indicates that recognition between the interacting partners occurs at several concurrent sites, which require the attainment of a defined spatial configuration, and highlights the possibility of compensation among many quasi-equivalent energetic contributions. Thus, the replacement of one or more energetically equivalent group(s) at the serine (pro)enzyme/BPTI (re)active site may affect only marginally the overall binding process. On the other hand, stereochemical constraint(s), and structural conservation within the serine protease superfamily, may account for the affinity displayed by BPTI for different homologous serine proteases (*e.g.*, b. b-trypsin versus h. a-thrombin) (see [9,11,12,15-17,19,89-91,104-105,122-124,129,135,136]).

From the dynamic point of view, the initial serine (pro)enzyme:BPTI encounter does not appear to be a smooth surface adhesion process leading to sequential formation of intermolecular interactions. Rather, it probably follows a groping process, until the system nestles in an energy minimum along the reaction pathway corresponding to the formation of the final inactive complex (see [9,11,12,15-17,19,89-91,104-105,122-124,129,135,136]).

As a whole, both structural and dynamic data concerning serine (pro)enzyme:BPTI complex (de)stabilization speak strongly in favor of Fink's proposal [189] that each of the observed effects subsequent to the initial (pro)enzyme:inhibitor or :substrate complex formation are indicative of subtle conformational changes that maximize interactions between the (pro)enzyme and the inhibitor or the substrate.

Clinically, BPTI exerts a significant hemostatic yet antithrombotic mechanism of action, which, together with its multitiered anti-inflammatory effect, makes this drug a valuable presidium in cardiac surgery and orthotopic liver transplantation. Thus, the use of BPTI in these clinical settings is advised, as it significantly reduces hemorrhagic complications, blood-transfusion requirements, and the inflammatory status associated to extra-corporeal circulation [13,20,23].

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ABBREVIATIONS

b.	=	Bovine
BPTI	=	Bovine basic pancreatic trypsin inhibitor (also named aprotinin, pancreatic Kunitz inhibitor, and trypsin-kallikrein inhibitor)
h.	=	Human
h. l.	=	Human leukocyte
h. p.	=	Human pancreatic
h. u.	=	Human urinary
KIU	=	Kallikrein inactivation units
NOS-I	=	Nitric oxide synthase type I
NOS-II	=	Nitric oxide synthase type II
p. p.	=	Porcine pancreatic

r. = Rat

Z-Lys-ONp = N- α -carbobenzoxyl-L-lysine *p*-nitrophenyl ester

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